OGC® DOCUMENT: YY-999

Open Geospatial Consortium

BEST PRACTICE FOR OGC GEODATACUBE MANAGEMENT

BEST PRACTICE
General

CANDIDATE SWG DRAFT

Version: 1.0
Submission Date: 2026-05-14
Approval Date: XXXX-XX-XX
Publication Date: XXXX-XX-XX
Editor: Alexander Jacob, Juraj Zvolenský, Yuvraj Bhagwan Adagale

Note

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. The Open Geospatial Consortium shall not be held responsible for identifying any or all such patent rights.

Recipients of this document are requested to submit, with their comments, notification of any relevant patent claims or other intellectual property rights of which they may be aware that might be infringed by any implementation of the standard set forth in this document, and to provide supporting documentation.

CONTENTS

I ABSTRACT

The rapidly growing volume of Earth observation data, together with the increasing diversity of data providers, platforms, formats, APIs, and processing environments, makes it difficult for users to consistently discover, access, prepare, and use data for analysis. While metadata standards and catalogues provide an essential foundation for discovery and description, additional guidance is needed on how discovered datasets can be transformed into analysis-ready geodatacubes and managed throughout their use.

This Best Practice provides recommendations for geodatacube management, focusing on the pathways from metadata and data discovery towards usable multidimensional datacubes. It describes how metadata-driven approaches can support the creation, configuration, access, and processing of geodatacubes across heterogeneous infrastructures. Building on previous work, including OGC Testbed 19, the document explores the role of openEO and relevant open-source EOEPCA building blocks as interoperable components for exposing, processing, and managing Earth observation data as geodatacubes.

The goal of this Best Practice is to provide practical guidance for data providers, platform operators, and application developers who need to design geodatacube management systems. It recommends approaches for bridging discovery metadata, data access mechanisms, processing interfaces, and datacube representations in a way that improves interoperability, reproducibility, and usability of Earth observation data.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, best practice, GeoDataCubes

III

PREFACE

NOTE: Insert Preface Text here. Give OGC specific commentary: describe the technical content, reason for document, history of the document and precursors, and plans for future work.

There are two ways to specify the Preface: “simple clause” or “full clasuse”

If the Preface does not contain subclauses, it is considered a simple preface clause. This one is entered as text after the `.Preface` label and must be placed between the AsciiDoc document attributes and the first AsciiDoc section title. It should not be give a section title of its own.

If the Preface contains subclauses, it needs to be encoded as a full preface clause. This one is recognized as a full Metanorma AsciiDoc section with te title “Preface”, i.e. `== Preface`. (Simple preface content can also be encoded like full preface.)

IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Eurac Research
- Organization two
- Organization three

VI SUBMITTERS

All questions regarding this submission should be directed to the editor or the submitters:

NAME	AFFILIATION	OGC MEMBER
Alexander Jacob	Eurac Research	Yes
Juraj Zvolenský	Eurac Research	Yes
Yuvraj Bhagwan Adagale	Eurac Research	Yes

VII CONTRIBUTORS

Individual name(s), Organization

NOTE: If you need to place any further sections in the preface area use the `[.preface]` attribute.

1

SCOPE

1 SCOPE

NOTE: This document specifies a set of best practice recommendations for the management of geospatial data as geodatacubes. The scope of this document is the transition from metadata-driven discovery of Earth observation datasets to their access, organization, processing, and use as multidimensional geodatacubes.

The document provides guidance on the use of metadata records, catalogues, data access information, and processing interfaces to support interoperable geodatacube management. It includes consideration of openEO as an interface for geodatacube access and processing, and considers the use of open-source EOEPCA building blocks as components within geodatacube management architectures.

This document is intended to support implementers of data platforms, processing services, and client applications that need to manage Earth observation data in a datacube-oriented form. It does not define a new standard for geodatacube encoding, metadata, or processing. Instead, it identifies recommended practices for combining existing standards, interfaces, and software components to support consistent and reusable geodatacube workflows.

2

CONFORMANCE

2 CONFORMANCE

This Best Practice defines XXXX.

Requirements for N standardization target types are considered:

- AAAA
- BBBB

Conformance with this Best Practice shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface Best Practice, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).
- Any one of the Distributed Computing Platform profiles specified in Annexes TBD through TBD (normative).

All requirements-classes and conformance-classes described in this document are owned by the documents(s) identified.

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Identification of Common Molecular Subsequences. Smith, T.F., Waterman, M.S., J. Mol. Biol. 147, 195–197 (1981)

ZIB Structure Prediction Pipeline: Composing a Complex Biological Workflow through Web Services. May, P., Ehrlich, H.C., Steinke, T. In: Nagel, W.E., Walter, W.V., Lehner, W. (eds.) Euro-Par 2006. LNCS, vol. 4128, pp. 1148–1158. Springer, Heidelberg (2006)

The Grid: Blueprint for a New Computing Infrastructure., Foster, I., Kesselman, C.. Morgan Kaufmann, San Francisco (1999).

Grid Information Services for Distributed Resource Sharing. Czajkowski, K., Fitzgerald, S., Foster, I., Kesselman, C. In: 10th IEEE International Symposium on High Performance Distributed Computing, pp. 181–184. IEEE Press, New York (2001)

The Physiology of the Grid: an Open Grid Services Architecture for Distributed Systems Integration. Foster, I., Kesselman, C., Nick, J., Tuecke, S. Technical report, Global Grid Forum (2002)

National Center for Biotechnology Information, http://www.ncbi.nlm.nih.gov

International Organization for Standardization (committee): ISO 19101-1:2014, *Geographic information — Reference model — Part 1: Fundamentals*. International Organization for Standardization, Geneva (2014). https://www.iso.org/standard/59164.html.

International Organization for Standardization (committee): ISO 19115-1:2014, *Geographic information — Metadata — Part 1: Fundamentals*. International Organization for Standardization, Geneva (2014). https://www.iso.org/standard/53798.html.

International Organization for Standardization (committee): ISO 19157:2013, *Geographic information — Data quality*. International Organization for Standardization, Geneva (2013). https://www.iso.org/standard/32575.html.

ISO: ISO 19139:2007, *Geographic information — Metadata — XML schema implementation*. ISO (2007).

International Organization for Standardization (committee): ISO 19115-3, *Geographic information — Metadata — Part 3: XML schema implementation for fundamental concepts*. International Organization for Standardization, Geneva https://www.iso.org/standard/80874.html.

Joan Masó, Lucy Bastin, Open Geospatial Consortium: OGC 15-097r1, *OGC® Geospatial User Feedback Standard: Conceptual Model*. Open Geospatial Consortium (2016). http://www.opengis.net/doc/IS/guf-conceptual/1.0.0.

Gerhard Gröger, Thomas H. Kolbe, Claus Nagel, Karl-Heinz Häfele, Open Geospatial Consortium: OGC 12-019, *OGC City Geography Markup Language (CityGML) Encoding Standard*. Open Geospatial Consortium (2012). http://www.opengis.net/spec/citygml/2.0.

Jiyeong Lee, Ki-Joune Li, Sisi Zlatanova, Thomas H. Kolbe, Claus Nagel, Thomas Becker, Open Geospatial Consortium: OGC 14-005r3, *OGC® IndoorGML*. Open Geospatial Consortium (2014). http://www.opengis.net/doc/IS/indoorgml/1.0.0.

Arliss Whiteside, Jim Greenwood, Open Geospatial Consortium: OGC 06-121r9, *OGC Web Service Common Implementation Specification*. Open Geospatial Consortium (2010). https://portal.ogc.org/files/?artifact_id=38867.

4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. example term

term used for exemplary purposes

Note 1 to entry: An example note.

Example Here’s an example of an example term.

[**SOURCE:** ISO 19101-1:2014]

6

CONVENTIONS

6 CONVENTIONS

This sections provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

6.1. Identifiers

The normative provisions in this document are denoted by the URI

`http://www.opengis.net/spec/{standard}/{m.n}`

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

6.2. Abbreviated terms

API Application Programming Interface

COG Cloud Optimized GeoTIFF

CWL Common Workflow Language

EO Earth Observation

EP Exploitation Platform

GDAL Geospatial Data Abstraction Library

JSON JavaScript Object Notation

OGC Open Geospatial Consortium

OS Operating System

REST Representational State Transfer

SAFE Standard Archive Format for Europe

SNAP Sentinel Application Platform toolbox

STAC SpatioTemporal Asset Catalog

TBD To Be Determined

TIFF Tagged Image File Format

URL Uniform Resource Locator

YAML YAML Ain’t Markup Language

7

INFORMATIVE OVERVIEW

7 INFORMATIVE OVERVIEW

This clause provides an informative overview of the role of a GeoDataCube Management System, the main architectural building blocks that can support such a system, and example usage scenarios. The purpose is to establish the context for the best practice recommendations provided later in this document.

The approach described in this Best Practice is based on the principle that a dedicated GeoDataCube-specific API is not required in order to manage and use geospatial data as datacubes. Instead, existing standards, APIs, metadata models, data access mechanisms, and processing interfaces can be combined as reusable building blocks. This building block approach supports interoperability while allowing implementations to select components that match their operational environment.

The scenarios and architectural considerations in this clause are loosely based on the GeoDataCube concepts explored in OGC Testbed 19 and are aligned with the initial objectives of the OGC GeoDataCube Standards Working Group. The emphasis is on practical pathways from metadata discovery to data access, datacube representation, processing, and publication of derived results.

7.1. GeoDataCube Management System

A GeoDataCube Management System is a set of services, software components, metadata records, data access mechanisms, and processing capabilities that allow users and applications to discover, access, organize, process, and reuse geospatial data as multidimensional datacubes.

In this document, a GeoDataCube Management System is understood as an integration pattern rather than as a single monolithic software system or API. Such a system can combine catalogue services, identity and access management, storage systems, data access interfaces, datacube access services, processing APIs, workspaces, and publication mechanisms.

A GeoDataCube Management System commonly supports the following high-level capabilities:

- discovery of relevant Earth observation and geospatial datasets through metadata catalogues;
- interpretation of metadata records and online assets as inputs to datacube creation;
- access to source data through machine-actionable links, APIs, or storage interfaces;
- representation of source data as virtual or materialized geodatacubes;
- execution of processing workflows over datacubes;
- publication of results with sufficient metadata and provenance to enable reuse;

- integration with authentication, authorization, workspace, and storage services.

The central concern of such a system is the transition from metadata to usable datacubes. Metadata records describe what data exists, where it is located, what it represents, and how it may be accessed. A GeoDataCube Management System uses that information to make the data available for analysis through datacube-oriented interfaces and processing environments.

7.2. From metadata to geodatacubes

Metadata catalogues are an essential starting point for geodatacube management. They enable users and systems to discover collections, products, scenes, assets, and derived results. However, discovery metadata alone is not sufficient for analysis. Additional information is required to determine how data can be accessed, combined, aligned, processed, and represented as a multidimensional datacube.

A typical metadata-to-geodatacube pathway includes the following steps:

1. discover a dataset, collection, or set of assets through a catalogue;
2. inspect metadata describing spatial extent, temporal extent, variables, bands, coordinate reference systems, resolution, format, licensing, and access constraints;
3. identify the online assets or service endpoints that provide access to the data;
4. determine whether the assets can be exposed directly as a virtual datacube or whether transformation is required;
5. apply any necessary harmonization, such as reprojection, resampling, band mapping, masking, or unit conversion;
6. expose the resulting logical datacube through a datacube access or processing interface;
7. execute processing workflows and capture provenance for derived outputs;
8. publish results back to a catalogue or workspace as reusable assets.

This pathway may be implemented in different ways depending on the underlying storage system, data format, processing platform, and user requirements. For example, some systems may expose cloud-optimized assets directly as virtual datacubes, while others may materialize data into chunked multidimensional formats such as Zarr or NetCDF for repeated access and scalable processing.

7.3. Reusable building block approach

Geodatacube management can be considered as an integration of reusable building blocks. Each building block addresses a specific concern, such as identity management, resource discovery, data access, processing, or workspace management. This approach avoids the need for a single GeoDataCube-specific API and allows implementations to build on existing OGC standards, community standards, and open-source software components.

The building block approach has the following advantages:

- it allows systems to reuse existing catalogue, access, and processing interfaces;
- it separates logical datacube management from physical storage formats;
- it supports multiple data access patterns, including direct object storage access, APIs, and service-based access;
- it allows processing interfaces such as openEO and OGC API — Processes to coexist;
- it supports federation across platforms and providers;
- it enables gradual adoption by existing infrastructures.

A building block may be implemented by different software components in different infrastructures. For example, resource discovery may be provided through OGC API — Records, STAC API, OpenSearch, or other catalogue interfaces. Processing may be provided through openEO, OGC API — Processes, Common Workflow Language execution environments, or platform-specific execution services.

7.4. EOEPCA building blocks

The Earth Observation Exploitation Platform Common Architecture (EOEPCA) provides an open-source reference architecture and set of building blocks for Earth observation exploitation platforms. These building blocks are relevant to GeoDataCube management because they address common platform capabilities required to move from data discovery to processing and reuse.

Table 1 — EOEPCA building blocks relevant to GeoDataCube management

BUILDING BLOCK	ROLE IN GEODATACUBE MANAGEMENT
Identity and Access Management	Provides authentication, authorization, user management, and access control for platform services, catalogues, data assets, processing resources, and workspaces.

BUILDING BLOCK	ROLE IN GEODATACUBE MANAGEMENT
Resource Discovery	Supports catalogue-based discovery of Earth observation data and related resources using interfaces such as OGC Catalogue Service for the Web, OGC API — Records, STAC, and OpenSearch.
Data Access	Provides interfaces to geospatial data assets stored or referenced by the platform. This may include access to object storage, online assets, APIs, file-based archives, or other data services.
Datacube Access	Allows users and applications to access and explore multidimensional Earth observation data through datacube-oriented access mechanisms and standard APIs.
openEO Processing API	Provides a datacube-oriented processing API for federated cloud processing of Earth observation data. It enables users to describe processing workflows using process graphs and execute them on compatible backends.
Workspace	Provides self-service environments for data access, algorithm development, collaborative exploration, and management of user or project-specific resources.

In addition to these building blocks, implementations may integrate workflow execution components based on OGC API — Processes, the Common Workflow Language (CWL), or other processing frameworks. Such components can complement openEO by supporting processing models that are not primarily datacube-oriented or that require packaging existing applications and command-line tools.

7.5. Role of openEO

openEO is a community standard and API for processing Earth observation data as datacubes. It provides a common abstraction for loading collections, filtering data in space and time, selecting bands, applying processes, executing batch jobs, and retrieving or publishing results.

Within a GeoDataCube Management System, openEO can serve as the primary user-facing processing interface for datacube workflows. It is particularly relevant where users need to interact with Earth observation data as multidimensional datacubes and where interoperability across processing backends is required.

The role of openEO in this Best Practice is not to replace metadata catalogues, storage systems, or all processing interfaces. Instead, openEO is positioned as a processing layer that can consume catalogue-described collections and expose them through a datacube processing model. Other processing approaches, including OGC API — Processes and CWL-based execution, may be used alongside openEO where appropriate.

openEO is particularly relevant in the European Earth observation ecosystem, where it is used or considered by major platforms and initiatives. Its adoption by strong user communities, including platforms such as Copernicus Data Space Ecosystem, Destination Earth, and related cloud processing environments, makes it a practical candidate for interoperable datacube processing.

7.6. Data formats, storage, and access patterns

GeoDataCube management systems need to operate across a wide range of formats, storage technologies, and access patterns. Source data may be provided as files, object storage assets, service endpoints, or platform-managed collections. Common examples include Cloud Optimized GeoTIFF, Zarr, NetCDF, SAFE archives, and other geospatial formats.

Cloud-optimized and chunked formats are particularly relevant for scalable datacube access. Cloud Optimized GeoTIFF can support efficient access to tiled raster assets over HTTP or object storage. Zarr can support multidimensional chunked arrays that are well suited to cloud-native datacube processing. NetCDF remains important for many scientific and atmospheric datasets.

A GeoDataCube Management System can avoid assuming that all data must be converted to a single physical format by distinguishing between the logical datacube presented to users and the physical layout of the source or derived assets. In some cases, a virtual datacube over existing assets is sufficient. In other cases, materializing a datacube into an optimized storage layout may be justified for performance, repeated access, machine learning workflows, or large-scale analysis.

Emerging access patterns, including artificial intelligence training workflows and Discrete Global Grid Systems, may place additional requirements on data layout, sampling, tiling, chunking, and metadata. These considerations can be addressed by implementation-specific profiles or operational guidance rather than by defining a new general-purpose GeoDataCube API.

7.7. Example usage scenarios

This section introduces three example scenarios that illustrate different levels of integration between discovery, access, processing, and workspace capabilities. These scenarios are informative and are not intended to prescribe a single architecture.

Table 2 — Summary of example GeoDataCube management scenarios

SCENARIO	BUILDING BLOCKS	DESCRIPTION
Simple data access	Identity and Access Management; Workspace; Resource Discovery	A user authenticates to a platform, opens a workspace, discovers relevant datasets through a catalogue, and accesses metadata and online assets for inspection or download.
Processing via openEO	Identity and Access Management; Workspace; Resource Discovery;	A user discovers a collection, loads it as a datacube through openEO, applies spatial and temporal filters,

SCENARIO	BUILDING BLOCKS	DESCRIPTION
	Data Access; Datacube Access; open EO Processing API	executes a processing graph, and retrieves or publishes the result.
Processing through application workflows	Identity and Access Management; Workspace; Resource Discovery; Data Access; Datacube Access; OGC API — Processes or CWL execution environment	A user discovers data and executes a packaged application or workflow, for example through OGC API — Processes or a CWL-based processing component, using catalogue-described assets and platform data access services.

7.7.1. Scenario 1: Simple data access

In the simplest scenario, the user interacts with a platform to discover and access data assets. The user authenticates through the Identity and Access Management component and uses a workspace or portal environment to search available resources through the Resource Discovery component.

The main objective of this scenario is to provide consistent discovery and access to metadata and online assets. The data may not yet be exposed through a datacube processing interface. However, the metadata can contain sufficient information to allow downstream systems or users to understand whether the data can be used as part of a geodatacube workflow.

A simplified component flow is:

```
Identity and Access Management -> Workspace -> Resource Discovery -> Online
assets
```

Listing 1

7.7.2. Scenario 2: Processing via openEO

In this scenario, catalogue-described datasets are made available through a datacube-oriented processing interface. The user authenticates, discovers relevant resources, and accesses the selected collection through an openEO backend. The openEO backend uses data access and datacube access components to load the required assets and expose them as a logical datacube.

The user can then define a process graph to filter, transform, aggregate, or otherwise process the datacube. The result may be downloaded, stored in a workspace, or published as a new catalogue resource with provenance metadata.

A simplified component flow is:

```
Identity and Access Management -> Workspace -> Resource Discovery -> Data Access
/ Datacube Access -> openEO Processing API
```

Listing 2

7.7.3. Scenario 3: Processing through OGC API — Processes or CWL

In this scenario, processing is performed using application-oriented workflows rather than, or in addition to, datacube-oriented openEO process graphs. The user discovers data through the catalogue and executes a packaged process or workflow through a processing component, such as an OGC API — Processes implementation or a CWL-based execution environment.

This scenario is useful where existing algorithms are already packaged as applications, command-line tools, containers, or workflow descriptions. It can complement openEO by supporting processing that does not naturally fit a datacube process graph or that requires specialized application execution.

A simplified component flow is:

```
Identity and Access Management -> Workspace -> Resource Discovery -> Data Access
/ Datacube Access -> OGC API - Processes / CWL execution
```

Listing 3

7.8. Relationship to best practice recommendations

The concepts and scenarios in this clause are informative. They establish the background for the recommendations in the following clauses. The normative part of this document can define which metadata, access, datacube description, processing, provenance, and publication practices are recommended for interoperable GeoDataCube management systems.

8

CLAUSE CONTAINING NORMATIVE MATERIAL

8 CLAUSE CONTAINING NORMATIVE MATERIAL

Paragraph

8.1. Requirement Class A or Requirement A Example

Paragraph – intro text for the requirement class.

Use the following table for Requirements Classes.

REQUIREMENTS CLASS 1	
OBLIGATION	requirement
DESCRIPTION	Requirements Class • http://www.example.org/req/blah • urn:iso:ts:iso:19139:clause:6
NORMATIVE STATEMENTS	Requirement 1-1 requirement description Requirement 1-2 requirement description Requirement 1-3 requirement description

8.1.1. Requirement 1

Paragraph — intro text for the requirement.

Use the following table for Requirements, number sequentially.

REQUIREMENT 1	
OBLIGATION	requirement
STATEMENT	Requirement 'shall' statement

Dictionary tables for requirements can be added as necessary. Modify the following example as needed.

Table 3

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name 1	definition of name 1	float	One or more (mandatory)
name 2	definition of name 2	character string type, not empty	Zero or one (optional)
name 3	definition of name 3	GML:: Point PropertyType	One (mandatory)

A

ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

NOTE: Ensure that there is a conformance class for each requirements class and a test for each requirement (identified by requirement name and number)

A.1. Conformance Class A

A.1.1. Requirement 1

REQUIREMENT A.1	
TEST PURPOSE	Verify that...
TEST METHOD	Inspect...

A.1.2. Requirement 2

B

ANNEX B (INFORMATIVE) TITLE ({NORMATIVE/INFORMATIVE})

B

ANNEX B (INFORMATIVE) TITLE ({NORMATIVE/INFORMATIVE})

NOTE: Place other Annex material in sequential annexes beginning with “B” and leave final two annexes for the Revision History and Bibliography

C

ANNEX C (INFORMATIVE) REVISION HISTORY

C ANNEX C (INFORMATIVE) REVISION HISTORY

Table C.1

DATE	RELEASE	EDITOR	PRIMARY CLAUSES MODIFIED	DESCRIPTION
2016-04-28	0.1	G. Editor	all	initial version

BIBLIOGRAPHY

BIBLIOGRAPHY

[1] *OGC: OGC Testbed 12 Annex B: Architecture* (2015).